WORKSHOP DECK

What predicts lexical decision speed?

One real lexical decision dataset

Regression as a tool for testing theory

Useful practice for later assessment work

RECAP

Recap: Lexical Decision

Participants decide whether a letter string is a **word or nonword**

Lower RT means **faster** recognition

Common words are usually recognised more quickly

Similar words can create lexical competition

DATASET

Meet the dataset

A prepared teaching version of a **public lexical decision megastudy**

Base source: **English Lexicon Project**

Each row is **one English word**

This is an **item analysis of word-level means**

VARIABLES

What the core variables mean

word
the lexical item

length
number of letters

log_frequency
how common the word is (**higher = more frequent**)

AoA
estimated age at which the word is learned (**higher = learned later**)

OLD20
a measure of orthographic neighbourhood similarity (**lower = more similar neighbours**)

mean_RT_ms
average lexical decision response time

KEY PREDICTOR

AoA

AoA stands for **Age of Acquisition**

Estimated age when a word is learned

Lower AoA = learned earlier; **higher AoA** = learned later

Earlier-acquired words are often recognised faster *(e.g. Morrison & Ellis, 2000)*

AoA can matter even after controlling for frequency

KEY PREDICTOR

OLD20

OLD20 is a measure of how similar a word is to other words in spelling

Based on the word's **20 closest spelling neighbours**

Lower OLD20 = more similar neighbours; **higher OLD20** = more distinct

Similar-looking words may increase lexical competition

More graded than a simple neighbour count

EXPLORE FIRST

First look at the data

Descriptives: individual variables

- Record summary statistics for **mean_RT_ms**, **length**, **log_frequency**, **AoA**, and **OLD20**
- What do the distributions look like?
- Are there missing values or implausible values?

Relationships: variables together

- Create scatterplots for key pairs of variables
- Start with **mean_RT_ms** against **log_frequency**
- Then look at relations between predictors
- Are there obvious trends, clusters, or outliers?

REGRESSION

Regression recap 1: what regression does

Regression tests whether one variable predicts another

It links a **dependent variable** to one or more **predictors**

It tells us about **direction**, **size**, and **uncertainty**

It moves us beyond simple description

WHY MULTIPLE REGRESSION

Regression recap 2: why multiple regression helps

Multiple regression includes several predictors in the same model

It shows whether each predictor explains **unique** variation

Coefficients are interpreted **holding other predictors constant**

This matters when predictors overlap

PREDICTOR OVERLAP

One simple relation between predictors

log_frequency ~ AoA

Start with one predictor-predictor relation

log_frequency with AoA

Questions:

- Are earlier-acquired words also more frequent?
  - Why does that matter in a multiple regression?

WORKSHOP

Building the models (stepwise)

- We'll build predictive models incrementally:
- | Model | Formula |
- | --- | --- |
- | Model 1 | **mean_RT_ms ~ log_frequency** |
- | Model 2 | **mean_RT_ms ~ log_frequency + length** |
- | Model 3 | **mean_RT_ms ~ log_frequency + length + AoA** |
- | Model 4 | **mean_RT_ms ~ log_frequency + length + AoA + OLD20** |

PREDICTION PROMPT

Now make your predictions

- Predict the direction of each relation with RT
- Higher **log_frequency** should predict...?
- Higher **AoA** should predict...?
- Greater **length** should predict...?
- Higher **OLD20** should predict...?
- Which predictors do you expect to remain once they are entered together?

CORE MODEL

Do these predictors explain unique RT variation?

mean_RT_ms ~ length + log_frequency + AoA + OLD20

Dependent variable: mean_RT_ms

Predictors: length, log_frequency, AoA, OLD20

Ask:

- Does the model explain RT variation?
  - Which predictors show unique associations?

INTERPRETATION

Turning coefficients into claims

For each predictor, note:

- coefficient sign
  - confidence interval or p value
  - plain-language interpretation

Ask:

- Do frequency and AoA both matter once the other predictors are included?

Better than "Which is strongest?":

- Which predictors show the clearest unique association?

Strength comparisons

If comparing relative strength, use **standardised estimates** only

LIMITATIONS

Common limitations of an item analysis

- **Word-level means** only
- Predictor overlap
- Correlational, not causal
- Hidden heterogeneity and **speed-accuracy trade-offs**

WRITTEN RESPONSE

What should your written response include?

- Key regression results
- Which hypotheses are supported
- A short plain-language interpretation

PAYOFF

What is the payoff?

Lexical decision RT is shaped by lexical properties

Regression helps us ask whether predictors make **unique** contributions

These predictors tell us something about the structure of the mental lexicon